

Argha Chakraborty

arghachakraborty18@gmail.com • +91 90885 35494 • Kolkata, India
linkedin.com/in/argha-chakraborty-359b02239 • github.com/argha-chakraborty • Google Scholar

Professional Summary

Computational Chemistry Ph.D. researcher with expertise in molecular dynamics simulations, statistical mechanics, and biomolecular self-assembly. Experienced in developing advanced simulation workflows using enhanced sampling techniques, free energy calculations, and machine learning-assisted molecular modeling. Proficient in OpenMM, NAMD, PLUMED, CHARMM, and Python-based scientific computing tools, with extensive experience working in Linux and high-performance computing environments.

Demonstrated track record of conducting independent research, developing novel computational methodologies, and analyzing large-scale molecular simulation datasets. Author of multiple peer-reviewed publications in leading chemistry journals, with research focused on understanding the kinetics, thermodynamics, and mechanistic pathways of peptide self-assembly at atomistic resolution. Recognized through national fellowships and multiple best poster presentation awards for contributions to theoretical and computational chemistry. Passionate about leveraging computational modeling and data-driven approaches to solve complex problems in chemical, biological, and materials sciences.

Core Technical Expertise

- **Simulation & Modeling:** Classical Molecular Dynamics, Enhanced Sampling and Free Energy Calculations, Simulation with machine learning interatomic potential, Ab initio molecular dynamics .
- **Software & Tools:** OpenMM, NAMD, PLUMED, CHARMM, VMD, PyMOL.
- **Programming & DevOps:** Python (NumPy, SciPy, PyTorch, MDTraj, MDAnalysis, Pandas), Bash Scripting, Linux/HPC Cluster Environment.

Education

Ph.D. in Chemistry 2022 – 2027 (Expected)

Indian Association for the Cultivation of Science, Kolkata, India

- Thesis Title: *Kinetic Mechanism and Free energy Landscape of Molecular Self-assembly*
- Advisor: [Dr. Avisek Das](#)

M.Sc. in Chemistry 2020 – 2022

Indian Association for the Cultivation of Science, Kolkata, India

B.Sc. in Chemistry (Honours) 2017 – 2020

St. Xavier's College, University of Calcutta, Kolkata, India

Research Experience

Senior Research Fellow 2022 – Present

School of Chemical Sciences, Indian Association for the Cultivation of Science

- Applied robust statistical mechanics frameworks to find out kinetics of micellization process
- Developed and deployed custom Python and OpenMM workflows utilizing **enhanced sampling techniques** (e.g., Umbrella Sampling, temperature accelerated MD, metadynamics) to explore complex, high-barrier conformational and aggregation states.
- Developed and algorithm named 'Event Tracking' to find out the pathway of any molecular self-assembly process.
- Automated structural and thermodynamic analyses of large-scale datasets using packages like MDAnalysis and MDTraj on high-performance computing (HPC) nodes.

Publications

1. How predictive is peptide self-assembly?: Correlating single- and bi-molecular structural properties with

nanoscale assemblies

Argha Chakraborty and Avisek Das*

Chemical Communications, (2026)

2. Atomistic picture of self-assembly dynamics of phenylalanine tripeptide into nanofibres: Hierarchical mechanism and the role of on-pathway water dynamics.
Rumela Adhikary, **Argha Chakraborty**, Sangeeta Das and Avisek Das*
Chemical Communications, 61, 16966 - 16969 (2025)
3. Understanding pathway complexity in assembly mechanisms of hydrophobic peptides in water: Insights from atomistic simulations via Event Tracking.
Argha Chakraborty, Rumela Adhikary, Sangeeta Das and Avisek Das*
The Journal of Chemical Physics, 162, 194712 (2025)
4. Correlating molecular structure and self-assembly mechanism via Temporal Analysis of Multidimensional Chemical Interactions Space: Understanding the difference between assembly behaviors of isomeric peptides in water.
Sangeeta Das, Rumela Adhikary, Snehamay Bagchi, **Argha Chakraborty** and Avisek Das*.
Physical Chemistry Chemical Physics, 27, 11264 (2025)

Teaching Experience

- Teaching assistant for masters level course 'Quantum Mechanics of Atoms and Molecules'
- Two times teaching assistant for masters level course 'Advanced Scientific Laboratory' for computer programming
- Teaching assistant for *National Supercomputing Mission* workshop for molecular simulation using GPU

Conferences & Presentations

- *Theoretical Chemistry Symposium*, 2025, Indian Institute of Technology, Mumbai.
- *Recent Advances in Modelling Rare Events*, 2025, Khajuraho, India
- *Interdisciplinary Symposium on Materials Chemistry*, 2024, Bhaba Atomic Research Centre, Mumbai
- *MD@60*, 2024, Jawaharlal Nehru Centre for Advanced Scientific Research, Bangalore.
- *Theoretical Chemistry Symposium*, 2023, Indian Institute of Technology, Madras.
- *Integrative multiscale modelling in biomolecular and soft matter systems*, 2023, Indian Institute of Science, Bangalore.

Awards & Fellowships

- Best Poster Presentation Award, *Interdisciplinary Symposium on Materials Chemistry*, 2024, Bhaba Atomic Research Centre, Mumbai
- Best Poster Presentation Award, *MD@60*, 2024, Jawaharlal Nehru Centre for Advanced Scientific Research, Bangalore.
- DST INSPIRE Scholarship, Department of Science and Technology, Government of India